

Real-Time Lab - Lab A Documentation and Data

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Files

Kernel Package files, cycletest outputs, and hello.c files can be found on [GitHub](#) in the realTimeLab folder under their respective folders. Other graphs, commands, and command outputs are shown in this PDF.

Step 0: Baseline capture

Baseline System Info

```
debian@beaglebone:~$ uname -a
Linux beaglebone 4.19.94-ti-r42 #1buster SMP PREEMPT Tue Mar 31 19:38:29
UTC 2020 armv7l GNU/Linux
debian@beaglebone:~$ cat /etc/os-release
PRETTY_NAME="Debian GNU/Linux 10 (buster)"
NAME="Debian GNU/Linux"
VERSION_ID="10"
VERSION="10 (buster)"
VERSION_CODENAME=buster
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
debian@beaglebone:~$ zcat /proc/config.gz | grep PREEMPT
# CONFIG_PREEMPT_NONE is not set
# CONFIG_PREEMPT_VOLUNTARY is not set
CONFIG_PREEMPT=y
CONFIG_PREEMPT_COUNT=y
CONFIG_PREEMPT_RCU=y
CONFIG_DEBUG_PREEMPT=y
# CONFIG_PREEMPTIRQ_EVENTS is not set
# CONFIG_PREEMPT_TRACER is not set
# CONFIG_PREEMPTIRQ_DELAY_TEST is not set
debian@beaglebone:~$ cat /proc/cmdline
console=ttyS0,115200n8 bone_capemgr.uboot_capemgr_enabled=1
root=/dev/mmcblk0p1 ro rootfstype=ext4 rootwait coherent_pool=1M
net.ifnames=0 lpj=1990656 rng_core.default_quality=100 quiet
```

Step 1: Patch and build a PREEMPT_RT kernel

1A. Install prerequisites

Using these links:

- <https://www.kernel.org/doc/html/v5.0/process/changes.html>
- <https://wiki.debian.org/BuildADebianKernelPackage>

We extract the required packages to build a kernel and write them into this command:

- compiler toolchain: build-essential
- make: make
- ncurses dev: libncurses5-dev
- flex: flex
- bison: bison
- OpenSSL dev: libssl-dev
- bc: bc
- common utilities: lzop, u-boot-tools, kmod, curl, wget, patch, cpio, fakeroot

```
sudo apt install build-essential make libncurses5-dev flex bison  
libssl-dev bc lzop u-boot-tools kmod curl wget patch cpio fakeroot -y
```

```
debian@beaglebone:~$ sudo apt install build-essential make libncurses5-dev  
flex bison libssl-dev bc lzop u-boot-tools kmod curl wget patch cpio  
fakeroot -y  
[sudo] password for debian:  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
bc is already the newest version (1.07.1-2+b1).  
build-essential is already the newest version (12.6).  
cpio is already the newest version (2.12+dfsg-9).  
kmod is already the newest version (26-1).  
lzop is already the newest version (1.03-4+b1).  
make is already the newest version (4.2.1-1.2).  
libncurses5-dev is already the newest version (6.1+20181013-2+deb10u2).  
patch is already the newest version (2.7.6-3+deb10u1).  
u-boot-tools is already the newest version (2019.01+dfsg-7).  
wget is already the newest version (1.20.1-1.1).  
The following additional packages will be installed:  
  libbison-dev libcurl4 libfakeroot libfl-dev libfl2 libssl1.1  
Suggested packages:  
  bison-doc flex-doc libssl-doc  
The following NEW packages will be installed:  
  bison fakeroot flex libbison-dev libfakeroot libfl-dev libfl2
```

The following packages will be upgraded:

curl libcurl4 libssl-dev libssl1.1

4 upgraded, 7 newly installed, 0 to remove and 179 not upgraded.

Need to get 5,592 kB of archives.

After this operation, 4,440 kB of additional disk space will be used.

Get:1 <http://archive.debian.org/debian buster/main armhf flex armhf>
2.6.4-6.2 [438 kB]

Get:2 <http://archive.debian.org/debian buster/main armhf libbison-dev armhf>
2:3.3.2.dfsg-1 [500 kB]

Get:3 <http://archive.debian.org/debian buster/main armhf bison armhf>
2:3.3.2.dfsg-1 [848 kB]

Get:4 <http://archive.debian.org/debian buster/main armhf libssl-dev armhf>
1.1.1n-0+deb10u3 [1,608 kB]

Get:5 <http://archive.debian.org/debian buster/main armhf libssl1.1 armhf>
1.1.1n-0+deb10u3 [1,310 kB]

Get:6 <http://archive.debian.org/debian buster/main armhf curl armhf>
7.64.0-4+deb10u2 [257 kB]

Get:7 <http://archive.debian.org/debian buster/main armhf libcurl4 armhf>
7.64.0-4+deb10u2 [298 kB]

Get:8 <http://archive.debian.org/debian buster/main armhf libfakeroot armhf>
1.23-1 [41.8 kB]

Get:9 <http://archive.debian.org/debian buster/main armhf fakeroot armhf>
1.23-1 [83.8 kB]

Get:10 <http://archive.debian.org/debian buster/main armhf libfl2 armhf>
2.6.4-6.2 [103 kB]

Get:11 <http://archive.debian.org/debian buster/main armhf libfl-dev armhf>
2.6.4-6.2 [104 kB]

Fetched 5,592 kB in 2s (2,682 kB/s)

Preconfiguring packages ...

Selecting previously unselected package flex.

(Reading database ... 72607 files and directories currently installed.)

Preparing to unpack .../00-flex_2.6.4-6.2_armhf.deb ...

Unpacking flex (2.6.4-6.2) ...

Selecting previously unselected package libbison-dev:armhf.

Preparing to unpack .../01-libbison-dev_2%3a3.3.2.dfsg-1_armhf.deb ...

Unpacking libbison-dev:armhf (2:3.3.2.dfsg-1) ...

Selecting previously unselected package bison.

Preparing to unpack .../02-bison_2%3a3.3.2.dfsg-1_armhf.deb ...

Unpacking bison (2:3.3.2.dfsg-1) ...

Preparing to unpack .../03-libssl-dev_1.1.1n-0+deb10u3_armhf.deb ...

Unpacking libssl-dev:armhf (1.1.1n-0+deb10u3) over (1.1.1d-0+deb10u2) ...

Preparing to unpack .../04-libssl1.1_1.1.1n-0+deb10u3_armhf.deb ...

Unpacking libssl1.1:armhf (1.1.1n-0+deb10u3) over (1.1.1d-0+deb10u2) ...

```

Preparing to unpack .../05-curl_7.64.0-4+deb10u2_armhf.deb ...
Unpacking curl (7.64.0-4+deb10u2) over (7.64.0-4+deb10u1) ...
Preparing to unpack .../06-libcurl4_7.64.0-4+deb10u2_armhf.deb ...
Unpacking libcurl4:armhf (7.64.0-4+deb10u2) over (7.64.0-4+deb10u1) ...
Selecting previously unselected package libfakeroot:armhf.
Preparing to unpack .../07-libfakeroot_1.23-1_armhf.deb ...
Unpacking libfakeroot:armhf (1.23-1) ...
Selecting previously unselected package fakeroot.
Preparing to unpack .../08-fakeroot_1.23-1_armhf.deb ...
Unpacking fakeroot (1.23-1) ...
Selecting previously unselected package libfl2:armhf.
Preparing to unpack .../09-libfl2_2.6.4-6.2_armhf.deb ...
Unpacking libfl2:armhf (2.6.4-6.2) ...
Selecting previously unselected package libfl-dev:armhf.
Preparing to unpack .../10-libfl-dev_2.6.4-6.2_armhf.deb ...
Unpacking libfl-dev:armhf (2.6.4-6.2) ...
Setting up flex (2.6.4-6.2) ...
Setting up libbison-dev:armhf (2:3.3.2.dfsg-1) ...
Setting up libssl1.1:armhf (1.1.1n-0+deb10u3) ...
Setting up libfakeroot:armhf (1.23-1) ...
Setting up fakeroot (1.23-1) ...
update-alternatives: using /usr/bin/fakeroot-sysv to provide
/usr/bin/fakeroot (
fakeroot) in auto mode
Setting up libssl-dev:armhf (1.1.1n-0+deb10u3) ...
Setting up libfl2:armhf (2.6.4-6.2) ...
Setting up bison (2:3.3.2.dfsg-1) ...
update-alternatives: using /usr/bin/bison.yacc to provide /usr/bin/yacc
(yacc) i
n auto mode
Setting up libcurl4:armhf (7.64.0-4+deb10u2) ...
Setting up curl (7.64.0-4+deb10u2) ...
Setting up libfl-dev:armhf (2.6.4-6.2) ...
Processing triggers for man-db (2.8.5-2) ...
Processing triggers for libc-bin (2.28-10) ...

```

1B. Select kernel source and matching RT patch

Currently Running

```
debian@beaglebone:~$ uname -r
```

4.19.94-ti-r42

Selecting kernel source version

Using the current version listed above, we can search for a similar version with: `apt search linux-image-4.19.94`

```
debian@beaglebone:~$ apt search linux-image-4.19.94
Sorting... Done
Full Text Search... Done
linux-image-4.19.94-armv7-lpae-x43/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-lpae-x43

linux-image-4.19.94-armv7-lpae-x44/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-lpae-x44

linux-image-4.19.94-armv7-rt-x43/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-rt-x43

linux-image-4.19.94-armv7-rt-x44/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-rt-x44

linux-image-4.19.94-armv7-x43/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-x43

linux-image-4.19.94-armv7-x44/unknown 1buster armhf
  Linux kernel, version 4.19.94-armv7-x44

linux-image-4.19.94-bone-rt-r43/unknown 1buster armhf
  Linux kernel, version 4.19.94-bone-rt-r43

linux-image-4.19.94-bone-rt-r44/unknown 1buster armhf
  Linux kernel, version 4.19.94-bone-rt-r44

linux-image-4.19.94-bone43/unknown 1buster armhf
  Linux kernel, version 4.19.94-bone43

linux-image-4.19.94-bone44/unknown 1buster armhf
  Linux kernel, version 4.19.94-bone44

linux-image-4.19.94-ti-r34/unknown 1buster armhf
  Linux kernel, version 4.19.94-ti-r34
```

```
linux-image-4.19.94-ti-r35/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r35

linux-image-4.19.94-ti-r36/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r36

linux-image-4.19.94-ti-r37/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r37

linux-image-4.19.94-ti-r38/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r38

linux-image-4.19.94-ti-r39/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r39

linux-image-4.19.94-ti-r40/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r40

linux-image-4.19.94-ti-r41/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r41

linux-image-4.19.94-ti-r42/unknown,now 1buster armhf [installed]
Linux kernel, version 4.19.94-ti-r42

linux-image-4.19.94-ti-r43/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r43

linux-image-4.19.94-ti-r44/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r44

linux-image-4.19.94-ti-r45/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r45

linux-image-4.19.94-ti-r47/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r47

linux-image-4.19.94-ti-r48/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r48

linux-image-4.19.94-ti-r49/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r49

linux-image-4.19.94-ti-r50/unknown 1buster armhf
```

Linux kernel, version [4.19.94-ti-r50](#)

linux-image-[4.19.94-ti-r51](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r51](#)

linux-image-[4.19.94-ti-r52](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r52](#)

linux-image-[4.19.94-ti-r53](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r53](#)

linux-image-[4.19.94-ti-r54](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r54](#)

linux-image-[4.19.94-ti-r55](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r55](#)

linux-image-[4.19.94-ti-r56](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r56](#)

linux-image-[4.19.94-ti-r57](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r57](#)

linux-image-[4.19.94-ti-r58](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r58](#)

linux-image-[4.19.94-ti-r59](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r59](#)

linux-image-[4.19.94-ti-r60](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r60](#)

linux-image-[4.19.94-ti-r61](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r61](#)

linux-image-[4.19.94-ti-r62](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r62](#)

linux-image-[4.19.94-ti-r63](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r63](#)

linux-image-[4.19.94-ti-r64](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-r64](#)

linux-image-4.19.94-ti-r65/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r65

linux-image-4.19.94-ti-r66/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r66

linux-image-4.19.94-ti-r67/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r67

linux-image-4.19.94-ti-r68/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r68

linux-image-4.19.94-ti-r69/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r69

linux-image-4.19.94-ti-r70/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r70

linux-image-4.19.94-ti-r71/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r71

linux-image-4.19.94-ti-r72/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r72

linux-image-4.19.94-ti-r73/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r73

linux-image-4.19.94-ti-r74/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-r74

linux-image-4.19.94-ti-rt-r34/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r34

linux-image-4.19.94-ti-rt-r35/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r35

linux-image-4.19.94-ti-rt-r36/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r36

linux-image-4.19.94-ti-rt-r37/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r37

linux-image-4.19.94-ti-rt-r38/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r38

linux-image-4.19.94-ti-rt-r39/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r39

linux-image-4.19.94-ti-rt-r40/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r40

linux-image-4.19.94-ti-rt-r41/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r41

linux-image-4.19.94-ti-rt-r42/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r42

linux-image-4.19.94-ti-rt-r43/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r43

linux-image-4.19.94-ti-rt-r44/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r44

linux-image-4.19.94-ti-rt-r45/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r45

linux-image-4.19.94-ti-rt-r47/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r47

linux-image-4.19.94-ti-rt-r48/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r48

linux-image-4.19.94-ti-rt-r49/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r49

linux-image-4.19.94-ti-rt-r50/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r50

linux-image-4.19.94-ti-rt-r51/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r51

linux-image-4.19.94-ti-rt-r52/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r52

linux-image-4.19.94-ti-rt-r53/unknown 1buster armhf

Linux kernel, version [4.19.94-ti-rt-r53](#)

linux-image-[4.19.94-ti-rt-r54](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r54](#)

linux-image-[4.19.94-ti-rt-r55](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r55](#)

linux-image-[4.19.94-ti-rt-r56](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r56](#)

linux-image-[4.19.94-ti-rt-r57](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r57](#)

linux-image-[4.19.94-ti-rt-r58](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r58](#)

linux-image-[4.19.94-ti-rt-r59](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r59](#)

linux-image-[4.19.94-ti-rt-r60](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r60](#)

linux-image-[4.19.94-ti-rt-r61](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r61](#)

linux-image-[4.19.94-ti-rt-r62](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r62](#)

linux-image-[4.19.94-ti-rt-r63](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r63](#)

linux-image-[4.19.94-ti-rt-r64](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r64](#)

linux-image-[4.19.94-ti-rt-r65](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r65](#)

linux-image-[4.19.94-ti-rt-r66](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r66](#)

linux-image-[4.19.94-ti-rt-r67](#)/unknown [1buster](#) armhf
Linux kernel, version [4.19.94-ti-rt-r67](#)

linux-image-4.19.94-ti-rt-r68/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r68

linux-image-4.19.94-ti-rt-r69/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r69

linux-image-4.19.94-ti-rt-r70/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r70

linux-image-4.19.94-ti-rt-r71/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r71

linux-image-4.19.94-ti-rt-r72/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r72

linux-image-4.19.94-ti-rt-r73/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r73

linux-image-4.19.94-ti-rt-r74/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r74

linux-image-4.19.94-ti-xenomai-r60/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r60

linux-image-4.19.94-ti-xenomai-r61/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r61

linux-image-4.19.94-ti-xenomai-r62/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r62

linux-image-4.19.94-ti-xenomai-r63/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r63

linux-image-4.19.94-ti-xenomai-r64/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r64

linux-image-4.19.94-ti-xenomai-r65/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r65

linux-image-4.19.94-ti-xenomai-r66/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-xenomai-r66

And will choose the version which matches the `4.19.94` value, containing `rt`, and matches the value of `r42` to ensure we select the correct kernel. This same kernel is shown in the results above as

```
linux-image-4.19.94-ti-rt-r42/unknown 1buster armhf
Linux kernel, version 4.19.94-ti-rt-r42
```

Thus, this leads us to install this kernel: `linux-image-4.19.94-ti-rt-r42`

1C. Configure from your running kernel

GitHub Clone

First trying to clone over the entire repo for the kernel, it failed due to the limited memory on the BBB:

```
debian@beaglebone:~$ git clone
https://github.com/RobertCNelson/linux-stable-rcn-ee
Cloning into 'linux-stable-rcn-ee'...
remote: Enumerating objects: 16899336, done.
remote: Counting objects: 100% (5151/5151), done.
remote: Compressing objects: 100% (1353/1353), done.
fatal: Out of memory, calloc failed
fatal: index-pack failed
```

Thus, I cloned the latest version of the specific kernel version I was looking for to then get:

```
debian@beaglebone:~$ git clone --depth 1 --branch 4.19.94-ti-rt-r42
https://github.com/RobertCNelson/linux-stable-rcn-ee
Cloning into 'linux-stable-rcn-ee'...
remote: Enumerating objects: 66099, done.
remote: Counting objects: 100% (66099/66099), done.
remote: Compressing objects: 100% (61801/61801), done.
remote: Total 66099 (delta 6074), reused 14200 (delta 3377), pack-reused 0
(from 0)
Receiving objects: 100% (66099/66099), 176.91 MiB | 927.00 KiB/s, done.
Resolving deltas: 100% (6074/6074), done.
Note: checking out 'dd627d9c27a85bbbc4ae65d99e9b40f8980bf3cb'.
```

You are in **'detached HEAD'** state. You can look around, make experimental changes **and** commit them, **and** you can discard any commits you make in **this** state without impacting any branches by performing another checkout.

If you want to create a **new** branch to retain commits you create, you may **do** so (now **or** later) by **using** **-b** with the checkout command again. Example:

```
git checkout -b <new-branch-name>
```

```
Checking out files: 100% (62262/62262), done.
```

Verify .config

```
debian@beaglebone:~/linux-stable-rcn-ee$ zcat /proc/config.gz > .config
debian@beaglebone:~/linux-stable-rcn-ee$ make olddefconfig
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  YACC    scripts/kconfig/zconf.tab.c
  LEX     scripts/kconfig/zconf.lex.c
  HOSTCC  scripts/kconfig/zconf.tab.o
  HOSTLD  scripts/kconfig/conf
scripts/kconfig/conf  --olddefconfig Kconfig
#
# configuration written to .config
#
debian@beaglebone:~/linux-stable-rcn-ee$ grep PREEMPT_RT .config
# CONFIG_PREEMPT_RT is not set
# CONFIG_PREEMPT_RT_FULL is not set
```

Then, realizing RT is not set, I used the script to update the RT flag

```
debian@beaglebone:~/linux-stable-rcn-ee$ ./scripts/config --enable
CONFIG_PREEMPT_RT_FULL
debian@beaglebone:~/linux-stable-rcn-ee$ grep PREEMPT_RT .config
# CONFIG_PREEMPT_RT is not set
CONFIG_PREEMPT_RT_FULL=y
```

1D. Build and install

Building Kernel

Using this link:

<https://forum.beagleboard.org/t/beagleboard-debian-how-to-download-kernel-source-packages/30524/3>

We found that the build command should include the `LOCALVERSION` flag, which in our case can be set to `-ti-rt-r42`, making our build command `make -j2 LOCALVERSION=-ti-rt-r42 bindeb-pkg` and accepted all default values for any flags that appeared during the building process:

```
debian@beaglebone:~/linux-stable-rcn-ee$ make -j2 LOCALVERSION=-ti-rt-r42
bindeb-pkg
scripts/kconfig/conf --synconfig Kconfig.config:86:warning: override:
PREEMPT+RT_FULL changes choice state
*
* Restarting config...
*
*
* RCU Subsystem
*
Make expert-level adjustments to RCU configuration (RCU_EXPERT) [N/y/?] n
Enable RCU priority boosting (RCU_BOOST) [Y/n/?] (NEW)
...
panic timeout (PANIC_TIMEOUT) [0] 0
Collect scheduler debugging info (SCHED_DEBUG) [Y/n/?] y
Collect scheduler statistics (SCHEDSTATS) [Y/n/?] y
Detect stack corruption on calls to schedule() (SCHED_STACK_END_CHECK)
[Y/n/?] y
Enable extra timekeeping sanity checking (DEBUG_TIMEKEEPING) [N/y/?] n
Debug preemptible kernel (DEBUG_PREEMPT) [Y/n/?] (NEW)
Stack backtrace support (STACKTRACE) [Y/?] y
Warn for all uses of unseeded randomness (WARN_ALL_UNSEEDED_RANDOM) [N/y/?] n
kobject debugging (DEBUG_KOBJECT) [N/y/?] n
Verbose BUG() reporting (adds 70K) (DEBUG_BUGVERBOSE) [Y/n/?] y
Debug linked list manipulation (DEBUG_LIST) [Y/?] y
Debug priority linked list manipulation (DEBUG_PI_LIST) [N/y/?] n
Debug SG table operations (DEBUG_SG) [N/y/?] n
Debug notifier call chains (DEBUG_NOTIFIERS) [N/y/?] n
Debug credential management (DEBUG_CREDENTIALS) [N/y/?] n
Force round-robin CPU selection for unbound work items
(DEBUG_WQ_FORCE_RR_CPU) [N/y/?] n
```

```

Force extended block device numbers and spread them (DEBUG_BLOCK_EXT_DEVT)
[N/y/?] n
Enable CPU hotplug state control (CPU_HOTPLUG_STATE_CONTROL) [N/y/?] n
Notifier error injection (NOTIFIER_ERROR_INJECTION) [M/n/y/?] m
    PM notifier error injection module (PM_NOTIFIER_ERROR_INJECT) [M/n/?] m
    OF reconfig notifier error injection module
(OFF_RECONFIG_NOTIFIER_ERROR_INJECT) [N/m/?] n
    Netdev notifier error injection module (NETDEV_NOTIFIER_ERROR_INJECT)
[N/m/?] n
Fault-injection framework (FAULT_INJECTION) [N/y/?] n
Latency measuring infrastructure (LATENCYTOP) [N/y/?] n
*
* Tracers
*
Tracers (FTRACE) [Y/n/?] y
    Kernel Function Tracer (FUNCTION_TRACER) [Y/?] y
    Kernel Function Graph Tracer (FUNCTION_GRAPH_TRACER) [Y/n/?] y
    Enable trace events for preempt and irq disable/enable (PREEMPTIRQ_EVENTS)
[N/y/?] n
    Interrupts-off Latency Tracer (IRQSOFF_TRACER) [N/y/?] n
    Preemption-off Latency Tracer (PREEMPT_TRACER) [N/y/?] (NEW)
...
INSTALL debian/headertmp/usr/include/linux/sunrpc/ (1 file)
INSTALL debian/headertmp/usr/include/linux/tc_act/ (15 files)
INSTALL debian/headertmp/usr/include/linux/tc_emark/ (5 files)
INSTALL debian/headertmp/usr/include/linux/usb/ (13 files)
INSTALL debian/headertmp/usr/include/linux/wimax/ (1 file)
INSTALL debian/headertmp/usr/include/asm/ (38 files)
dpkg-deb: building package 'linux-headers-4.19.94-ti-rt-r42' in
'../linux-headers-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb'.
dpkg-deb: building package 'linux-libc-dev' in
'../linux-libc-dev_4.19.94-ti-rt-r42-1_armhf.deb'.
dpkg-deb: building package 'linux-image-4.19.94-ti-rt-r42' in
'../linux-image-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb'.
dpkg-genbuildinfo --build=binary
dpkg-genchanges --build=binary
>../linux-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.changes
dpkg-genchanges: warning: package linux-image-4.19.94-ti-rt-r42-dbg in
control file but not in files list
dpkg-genchanges: info: binary-only upload (no source code included)
dpkg-source --after-build .
dpkg-buildpackage: info: binary-only upload (no source included)
debian@beaglebone:~$ sudo dpkg -i *.deb

```

```
[sudo] password for debian:
Selecting previously unselected package linux-headers-4.19.94-ti-rt-r42.
(Reading database ... 72899 files and directories currently installed.)
Preparing to unpack
linux-headers-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb ...
Unpacking linux-headers-4.19.94-ti-rt-r42 (4.19.94-ti-rt-r42-1) ...
Selecting previously unselected package linux-image-4.19.94-ti-rt-r42.
Preparing to unpack
linux-image-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb ...
Unpacking linux-image-4.19.94-ti-rt-r42 (4.19.94-ti-rt-r42-1) ...
dpkg: warning: downgrading linux-libc-dev:armhf from 4.19.98-1 to
4.19.94-ti-rt-r42-1
Preparing to unpack linux-libc-dev_4.19.94-ti-rt-r42-1_armhf.deb ...
Unpacking linux-libc-dev (4.19.94-ti-rt-r42-1) over (4.19.98-1) ...
Setting up linux-headers-4.19.94-ti-rt-r42 (4.19.94-ti-rt-r42-1) ...
Setting up linux-image-4.19.94-ti-rt-r42 (4.19.94-ti-rt-r42-1) ...
update-initramfs: Generating /boot/initrd.img-4.19.94-ti-rt-r42
zz-uenv_txt: Updating /boot/uEnv.txt [uname_r=4.19.94-ti-rt-r42]
Setting up linux-libc-dev (4.19.94-ti-rt-r42-1) ...
```

1E. Verify after reboot (required proof)

```
debian@beaglebone:~$ sudo reboot
debian@beaglebone:~$ Connection to 192.168.7.2 closed by remote host.
Connection to 192.168.7.2 closed.
k@k:~$ ssh debian@192.168.7.2
Debian GNU/Linux 10
```

BeagleBoard.org Debian Buster IoT Image [2020-04-06](#)

Support: http://elinux.org/Beagleboard:BeagleBoneBlack_Debian

default username:password is [debian:temppwd]

debian@192.168.7.2's password:

The programs included with the Debian GNU/Linux system are **free** software; the exact distribution terms **for** each program are described in the individual files in `/usr/share/doc/*/copyright`.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.


```
Last Login: Sat Apr  4 08:46:29 2026 from 192.168.7.1
debian@beaglebone:~$ uname -a
Linux beaglebone 4.19.94-ti-rt-r42 #1 SMP PREEMPT RT Fri Apr 3 21:34:09 UTC
2026 armv7l GNU/Linux
```

Step 2: “Hello world” on the RT kernel

Compile and run a simple C program

Program

```
#include <stdint.h>
#include <stdio.h>
#include <sys/utsname.h>

int main() {
    struct utsname sys_info = { 0 };
    int32_t result = uname(&sys_info);
    printf("System information: %s, %s, %s, %s\n", sys_info.sysname,
sys_info.release, sys_info.version, sys_info.machine);
    printf("hello from BBB\n");
    return 0;
}
```

Output

```
debian@beaglebone:~$ gcc -o hello hello_bbb.c
debian@beaglebone:~$ ./hello
System information: Linux, 4.19.94-ti-rt-r42, #1 SMP PREEMPT RT Fri Apr 3
21:34:09 UTC 2026, armv7l
hello from BBB
```

Step 3:

Non-PREEMPT_RT Kernel

3A. Install rt-tests

```
debian@beaglebone:~$ sudo apt-get install rt-tests stress -y
Reading package lists... Done
Building dependency tree
```

```
Reading state information... Done
The following NEW packages will be installed:
  rt-tests stress
0 upgraded, 2 newly installed, 0 to remove and 183 not upgraded.
Need to get 106 kB of archives.
After this operation, 314 kB of additional disk space will be used.
Get:1 http://archive.debian.org/debian buster/main armhf rt-tests armhf 1.0-3 [83.0 kB]
Get:2 http://archive.debian.org/debian buster/main armhf stress armhf 1.0.4-4 [22.6 kB]
Fetched 106 kB in 0s (325 kB/s)
Selecting previously unselected package rt-tests.
(Reading database ... 72570 files and directories currently installed.)
Preparing to unpack .../rt-tests_1.0-3_armhf.deb ...
Unpacking rt-tests (1.0-3) ...
Selecting previously unselected package stress.
Preparing to unpack .../stress_1.0.4-4_armhf.deb ...
Unpacking stress (1.0.4-4) ...
Setting up stress (1.0.4-4) ...
Setting up rt-tests (1.0-3) ...
Processing triggers for man-db (2.8.5-2) ...
```

3B. Reduce measurement noise

```
debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
performance
debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_available_governors
conservative ondemand userspace powersave performance schedutil
debian@beaglebone:~$ sudo sh -c 'echo performance >
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor'
debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
performance
```

3C. Collect data in two conditions

Idle-ish

```
debian@beaglebone:~$ cyclictest --help
cyclictest V 1.00
Usage:
cyclictest <options>
```

-a [NUM] --affinity run thread #N on processor #N, if possible
 with NUM pin all threads to the processor NUM
 -A USEC --aligned=USEC align thread wakeups to a specific offset
 -b USEC --breaktrace=USEC send **break** trace command when latency > USEC
 -B --preemptirqs both preempt **and** irqsoff tracing (used with -b)
 -c CLOCK --clock=CLOCK select clock
 0 = CLOCK_MONOTONIC (default)
 1 = CLOCK_REALTIME
 -C --context context **switch** tracing (used with -b)
 -d DIST --distance=DIST distance of thread intervals in us, **default=500**
 -D --duration=TIME specify a length **for** the test run.
 Append 'm', 'h', or 'd' to specify minutes,
 hours **or** days.
 --latency=PM_QOS write PM_QOS to /dev/cpu_dma_latency
 -E --event event tracing (used with -b)
 -f --ftrace function trace (when -b is active)
 -F --fifo=<path> create a named pipe at path **and** write stats to
 it
 -h --histogram=US dump a latency histogram to **stdout** after the run
 US is the max latency time to be tracked in
 microseconds
 This option runs all threads at the same priority.
 -H --histofall=US same as -h except with an additional summary
 column
 --histfile=<path> dump the latency histogram to <path> instead of
stdout
 -i INTV --interval=INTV base interval of thread in us **default=1000**
 -I --irqsoff Irqsoff tracing (used with -b)
 -l LOOPS --loops=LOOPS number of loops: **default=0**(endless)
 --laptop Save battery when running cyclicttest
 This will give you poorer realtime results
 but will **not** drain your battery so quickly
 -m --mlockall lock current **and** future memory allocations
 -M --refresh_on_max delay updating the screen until a **new** max
 latency is hit. Useful **for** low bandwidth.
 -n --nanosleep use clock_nanosleep
 --notrace suppress tracing
 -N --nsecs print results in ns instead of us (**default** us)
 -o RED --oscope=RED oscilloscope mode, reduce verbose output by RED
 -O TOPT --traceopt=TOPT trace option
 -p PRIO --priority=PRIO priority of highest prio thread
 -P --preemptoff Preempt off tracing (used with -b)
 --policy=NAME policy of measurement thread, where NAME may be one

```

of: other, normal, batch, idle, fifo or rr.
--priospread    spread priority levels starting at specified value
-q             --quiet        print a summary only on exit
-r             --relative     use relative timer instead of absolute
-R             --resolution   check clock resolution, calling clock_gettime()
many
times. List of clock_gettime() values will be
reported with -X
--secaligned [USEC] align thread wakeups to the next full second
and apply the optional offset
-s             --system       use sys_nanosleep and sys_setitimer
-S             --smp          Standard SMP testing: options -a -t -n and
same priority of all threads
--spike=<trigger> record all spikes > trigger
--spike-nodes=[num of nodes]
These are the maximum number of spikes we can record.
The default is 1024 if not specified
-t             --threads      one thread per available processor
-t [NUM] --threads=NUM       number of threads:
without NUM, threads = max_cpus
without -t default = 1
--tracemark     write a trace mark when -b latency is exceeded
-T TRACE --tracer=TRACER    set tracing function
configured tracers: none
-u             --unbuffered    force unbuffered output for live processing
-v             --verbose       output values on stdout for statistics
format: n:c:v n=tasknum c=count v=value in us
-w             --wakeup        task wakeup tracing (used with -b)
-W             --wakeuprt      rt task wakeup tracing (used with -b)
--dbg_cyclictest print info useful for debugging cyclictest
debian@beaglebone:~$ sudo cyclictest -a 0 -h 400 -i 1000 -D 1m -m -n -p 90
-q > baseline_idle_latency.txt
[sudo] password for debian:
debian@beaglebone:~$ ls
baseline_idle_latency.txt

```

```

-a 0: runs thread on first processor
-h 400: used for latency histogram
-i 1000: base interval of thread
-D 1m: duration for run
-m: locks memory allocations
-n: uses clock_nanosleep
-p 90: sets priority to 90 (high priority)
-q: print summary at completion of command

```

Loaded

CPU load with built-in tools using the stress package

```
debian@beaglebone:~$ stress --help
`stress' imposes certain types of compute stress on your system
```

Usage: stress [OPTION [ARG]] ...

-?, --help	show this help statement
--version	show version statement
-v, --verbose	be verbose
-q, --quiet	be quiet
-n, --dry-run	show what would have been done
-t, --timeout N	timeout after N seconds
--backoff N	wait factor of N microseconds before work starts
-c, --cpu N	spawn N workers spinning on sqrt()
-i, --io N	spawn N workers spinning on sync()
-m, --vm N	spawn N workers spinning on malloc()/free()
--vm-bytes B	malloc B bytes per vm worker (default is 256MB)
--vm-stride B	touch a byte every B bytes (default is 4096)
--vm-hang N	sleep N secs before free (default none, 0 is inf)
--vm-keep	redirty memory instead of freeing and reallocating
-d, --hdd N	spawn N workers spinning on write()/unlink()
--hdd-bytes B	write B bytes per hdd worker (default is 1GB)

Example: stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 10s

Note: Numbers may be suffixed with s,m,h,d,**y** (time) or B,K,M,**G** (size).

```
debian@beaglebone:~$ stress -c 4 -i 2 -m 1 --vm-bytes 128M -t 70 &
[1] 16928
debian@beaglebone:~$ stress: info: [16928] dispatching hogs: 4 cpu, 2 io, 1
vm, 0 hdd
stress -c 4 -i 2 -m 1sudo cyclictest -a 0 -h 400 -i 1000 -D 1m -m -n -p 90
-q > baseline_loaded_latency.txt
stress: info: [16928] successful run completed in 70s
[1]+  Done                  stress -c 4 -i 2 -m 1 --vm-bytes 128M -t 70
```

-c 4: 4 sqrt() workers

-i 4 sync() workers

-m 1: 1 malloc/free worker

--vm-bytes: 128MB as 512 RAM is on BBB

-t 70: timeout after 70 sec to have it close after cyclictest command is done

PREEMPT_RT Kernel

3A. Install rt-tests

```
debian@beaglebone:~$ sudo apt-get install rt-tests stress -y
[sudo] password for debian:
Reading package lists... Done
Building dependency tree
Reading state information... Done
rt-tests is already the newest version (1.0-3).
stress is already the newest version (1.0.4-4).
0 upgraded, 0 newly installed, 0 to remove and 179 not upgraded.
```

3B. Reduce measurement noise

```

debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
performance
debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_available_governors
conservative ondemand userspace powersave performance schedutil
debian@beaglebone:~$ sudo sh -c 'echo performance >
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor'
debian@beaglebone:~$ cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
performance

```

3C. Collect data in two conditions

Idle-ish

```
debian@beaglebone:~$ cyclictest --help
cyclictest V 1.00
Usage:
cyclictest <options>
```

[illegible]

```

1 = CLOCK_REALTIME
-C      --context      context switch tracing (used with -b)
-d DIST --distance=DIST distance of thread intervals in us, default=500
-D      --duration=TIME specify a length for the test run.
                        Append 'm', 'h', or 'd' to specify minutes,
hours or days.
      --latency=PM_QOS write PM_QOS to /dev/cpu_dma_latency
-E      --event        event tracing (used with -b)
-f      --ftrace       function trace (when -b is active)
-F      --fifo=<path>   create a named pipe at path and write stats to
it
-h      --histogram=US  dump a latency histogram to stdout after the run
                        US is the max latency time to be tracked in
microseconds
                        This option runs all threads at the same priority.
-H      --histofall=US  same as -h except with an additional summary
column
      --histfile=<path> dump the latency histogram to <path> instead of
stdout
-i INTV --interval=INTV base interval of thread in us default=1000
-I      --irqsoff       Irqsoff tracing (used with -b)
-l LOOPS --loops=LOOPS  number of loops: default=0(endless)
      --laptop          Save battery when running cyclictest
                        This will give you poorer realtime results
                        but will not drain your battery so quickly
-m      --mlockall      lock current and future memory allocations
-M      --refresh_on_max delay updating the screen until a new max
                        latency is hit. Useful for low bandwidth.
-n      --nanosleep     use clock_nanosleep
      --notrace         suppress tracing
-N      --nsecs         print results in ns instead of us (default us)
-o RED  --oscope=RED     oscilloscope mode, reduce verbose output by RED
-O TOPT --traceopt=TOPT trace option
-p PRIO --priority=PRIO priority of highest prio thread
-P      --preemptoff    Preempt off tracing (used with -b)
      --policy=NAME     policy of measurement thread, where NAME may be one
                        of: other, normal, batch, idle, fifo or rr.
      --priospread      spread priority levels starting at specified value
-q      --quiet         print a summary only on exit
-r      --relative      use relative timer instead of absolute
-R      --resolution    check clock resolution, calling clock_gettime()
many
                        times. List of clock_gettime() values will be

```

```

                                reported with -X
--secaligned [USEC] align thread wakeups to the next full second
                                and apply the optional offset
-s      --system                use sys_nanosleep and sys_setitimer
-S      --smp                   Standard SMP testing: options -a -t -n and
                                same priority of all threads
--spike=<trigger> record all spikes > trigger
--spike-nodes=[num of nodes]

```

These are the maximum number of spikes we can record.

The **default** is **1024** if not specified

```

-t      --threads                one thread per available processor
-t [NUM] --threads=NUM          number of threads:
                                without NUM, threads = max_cpus
                                without -t default = 1
                                --tracemark write a trace mark when -b latency is exceeded
-T TRACE --tracer=TRACER set tracing function
                                configured tracers: none
-u      --unbuffered            force unbuffered output for live processing
-v      --verbose                output values on stdout for statistics
                                format: n:c:v n=tasknum c=count v=value in us
-w      --wakeup                task wakeup tracing (used with -b)
-W      --wakeuprt              rt task wakeup tracing (used with -b)
--dbg_cyclictest print info useful for debugging cyclictest

```

```

debian@beaglebone:~$ sudo cyclictest -a 0 -h 400 -i 1000 -D 1m -m -n -p 90

```

```

-q > preempt_rt_idle_latency.txt

```

```

debian@beaglebone:~$ ls

```

```

baseline_idle_latency.txt

```

```

linux-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.changes

```

```

baseline_loaded_latency.txt

```

```

linux-headers-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb

```

```

bin

```

```

linux-image-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.deb

```

```

hello

```

```

linux-libc-dev_4.19.94-ti-rt-r42-1_armhf.deb

```

```

hello_bbb.c

```

```

linux-stable-rcn-ee

```

```

linux-4.19.94-ti-rt-r42_4.19.94-ti-rt-r42-1_armhf.buildinfo

```

```

preempt_rt_idle_latency.txt

```

-a 0: runs thread on first processor

-h 400: used for latency histogram

-i 1000: base interval of thread

-D 1m: duration for run

-m: locks memory allocations

-n: uses clock_nanosleep
-p 90: sets priority to 90 (high priority)
-q: print summary at completion of command

Loaded

CPU load with built-in tools using the stress package

```
debian@beaglebone:~$ stress --help
`stress' imposes certain types of compute stress on your system
```

Usage: stress [OPTION [ARG]] ...

-?, --help	show this help statement
--version	show version statement
-v, --verbose	be verbose
-q, --quiet	be quiet
-n, --dry-run	show what would have been done
-t, --timeout N	timeout after N seconds
--backoff N	wait factor of N microseconds before work starts
-c, --cpu N	spawn N workers spinning on sqrt()
-i, --io N	spawn N workers spinning on sync()
-m, --vm N	spawn N workers spinning on malloc()/free()
--vm-bytes B	malloc B bytes per vm worker (default is 256MB)
--vm-stride B	touch a byte every B bytes (default is 4096)
--vm-hang N	sleep N secs before free (default none, 0 is inf)
--vm-keep	redirty memory instead of freeing and reallocating
-d, --hdd N	spawn N workers spinning on write()/unlink()
--hdd-bytes B	write B bytes per hdd worker (default is 1GB)

Example: stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 10s

Note: Numbers may be suffixed with s,m,h,d,**y** (time) or B,K,M,**G** (size).

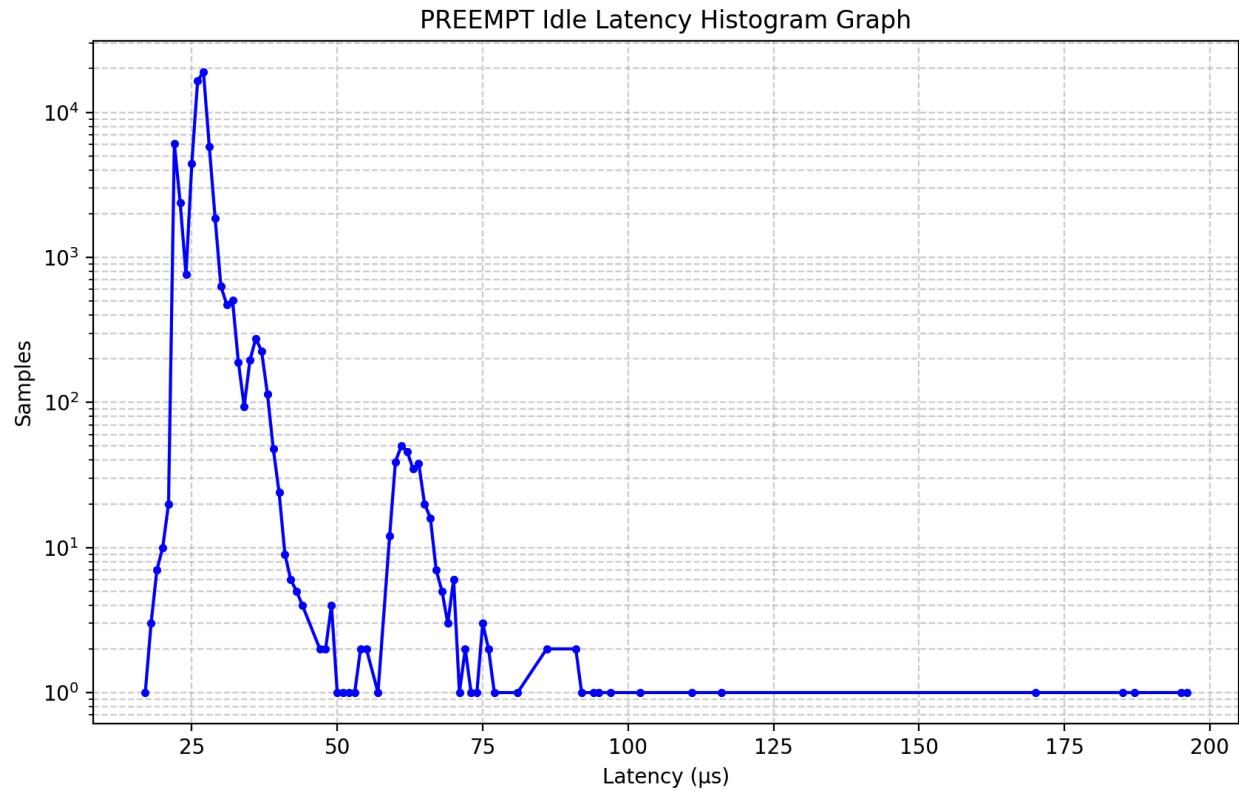
```
debian@beaglebone:~$ stress -c 4 -i 2 -m 1 --vm-bytes 128M -t 70 &
[1] 1815
debian@beaglebone:~$ stress: info: [1815] dispatching hogs: 4 cpu, 2 io, 1
vm, 0 hdd
sudo cyclictst -a 0 -h 400 -i 1000 -D 1m -m -n -p 90 -q >
preempt_rt_loaded_latency.txt
debian@beaglebone:~$ stress: info: [1815] successful run completed in 70s
```

-c 4: 4 sqrt() workers
-i 4 sync() workers
-m 1: 1 malloc/free worker
-vm-bytes: 128MB as 512 RAM is on BBB

-t 70: timeout after 70 sec to have it close after cyclicttest command is done

3D. Graph and reflect

PREEMPT Idle Graph



Total: 000060000

Min Latencies: 00017

Avg Latencies: 00026

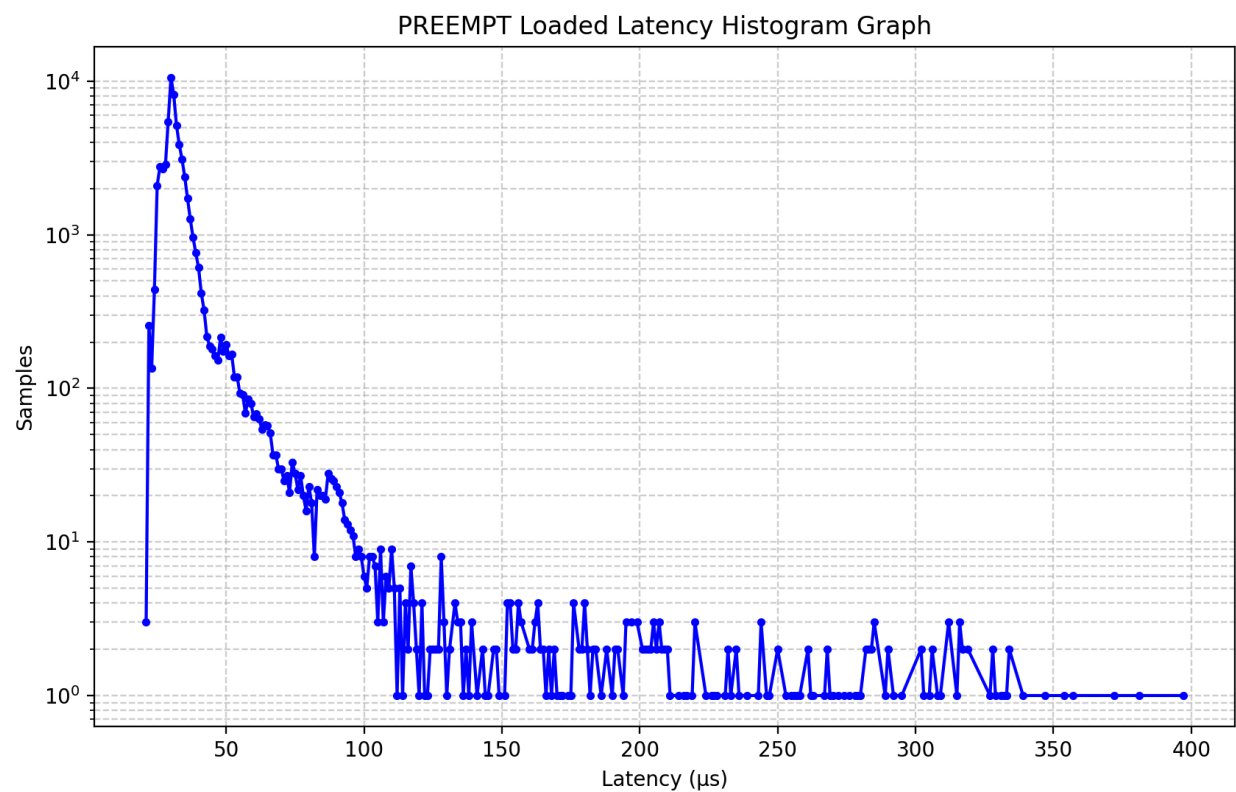
Max Latencies: 00196

Histogram Overflows: 00000

Histogram Overflow at cycle number:

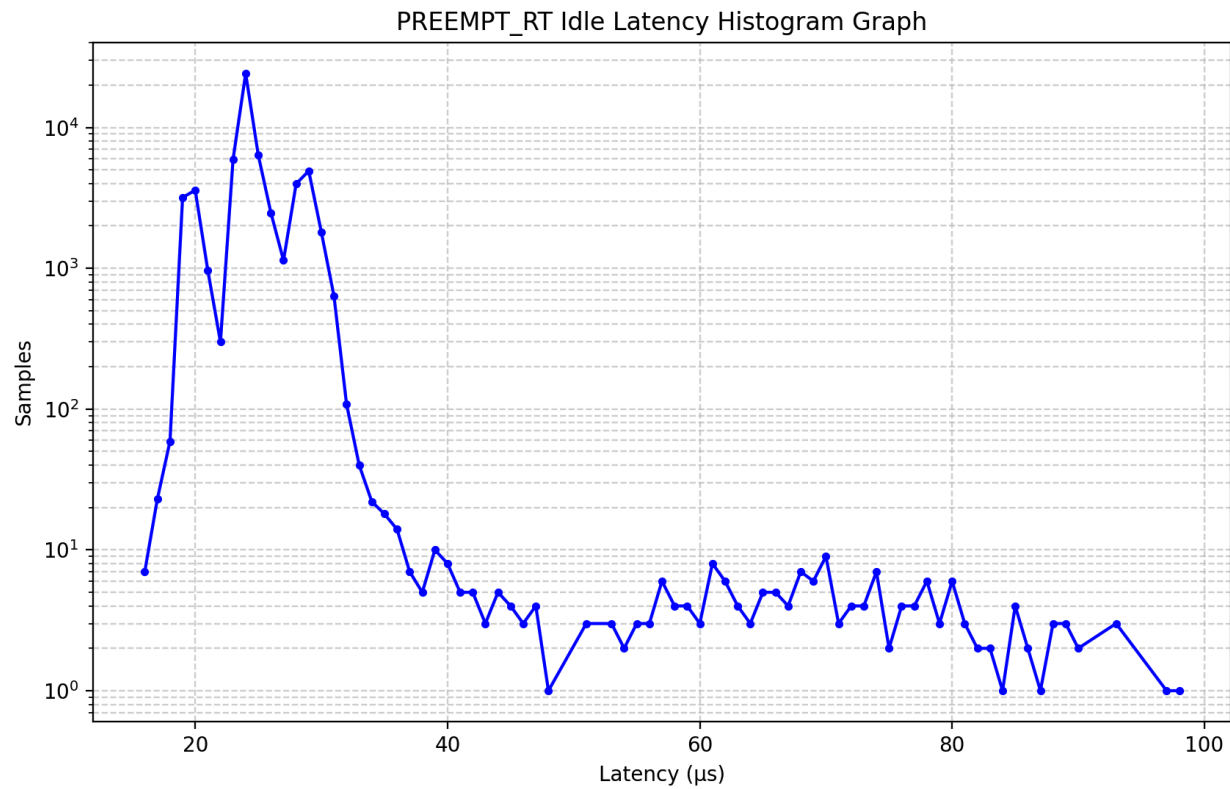
Thread 0:

PREEMPT Loaded Graph



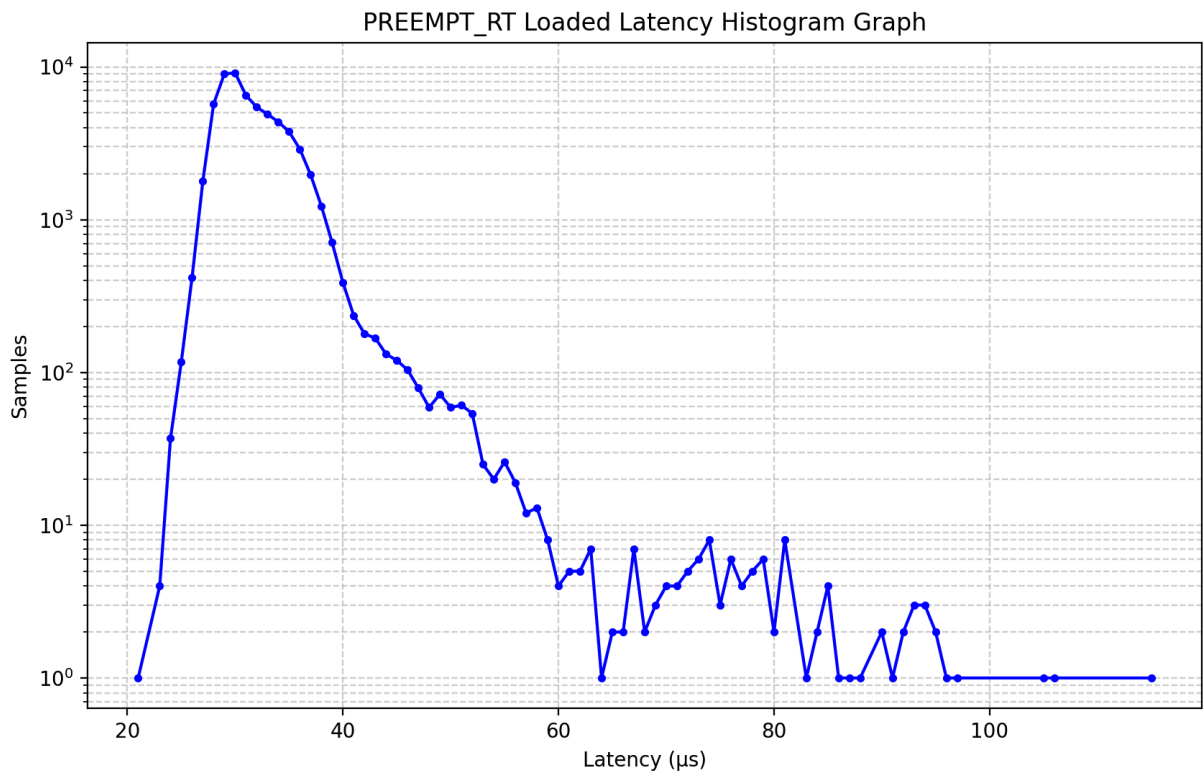
Total: 000060000
Min Latencies: 00021
Avg Latencies: 00033
Max Latencies: 00397
Histogram Overflows: 00000
Histogram Overflow at cycle number:
Thread 0:

PREEMPT_RT Idle Graph



Total: 000060000
Min Latencies: 00016
Avg Latencies: 00024
Max Latencies: 00098
Histogram Overflows: 00000
Histogram Overflow at cycle number:
Thread 0:

PREEMPT_RT Loaded Graph



Total: 000060000
Min Latencies: 00021
Avg Latencies: 00032
Max Latencies: 00115
Histogram Overflows: 00000
Histogram Overflow at cycle number:
Thread 0:

Latencies Table

Kernel Type/Load	Min Latencies	Avg Latencies	Max Latencies
PREEMPT Idle	00017	00026	00196
PREEMPT Loaded	00021	00033	00397
PREEMPT_RT Idle	00016	00024	00098
PREEMPT_RT Loaded	00021	00032	00115

Reflection

Using the latency values for both kernel and load types, the overall trend shown is that, in all cases, the PREEMPT_RT kernel achieved lower or the same min, avg, or max latency than the base PREEMPT kernel. The largest disparities are shown through the PREEMPT kernel vs PREEMPT_RT kernel when under an intentional CPU load, where the value dropped from 397 μ s to 115 μ s. Following in a similar set of footprints were the same set of kernels under an idle CPU load, where we were still able to see that the PREEMPT_RT kernel had a max latency of 98 μ s, vs the PREEMPT kernel had a max latency of 196 μ s. Smaller gaps can be seen when looking at average latencies under both loads, where the PREEMPT kernel under an idle load had an average latency of 26 μ s, vs the PREEMPT_RT kernel under an idle load had 24 μ s. Likewise, the PREEMPT kernel under an intentional CPU load performed with an average latency of 33 μ s, while the PREEMPT_RT kernel performed with a latency of 32 μ s. Moving to the last set, we can see that regardless of which kernel type was running, the intentional load had a minimum latency of 21 μ s, but while testing the idle CPU load, the PREEMPT_RT kernel outperformed the PREEMPT kernel by 1 μ s (16 μ s vs 17 μ s).

Looking particularly at the graphs for an intentionally loaded CPU, the PREEMPT_RT graph shows that it sampled the peak of its tests within 60 μ s each, whereas the PREEMPT kernel took around 100 μ s. Furthermore, there were numerous samples left over, taking more than 100 μ s in the PREEMPT kernel compared to the PREEMPT_RT kernel, which completed most of its samples within less than 100 μ s each. Comparing the PREEMPT_RT and PREEMPT kernel graphs under an idle CPU load, the PREEMPT kernel experienced two larger spikes, whereas PREEMPT_RT showed a single large (but slightly wider) spike. Additionally, looking at the volume of samples tested with respect to the x-axis, most of the PREEMPT_RT kernel samples still occurred before the second peak found in the baseline kernel. This second spike found in the PREEMPT_RT idle CPU load could be due to advantages the PREEMPT_RT kernel has compared to the baseline kernel, such as determinism and hard real-time requirements. However, even with the use of the PREEMPT_RT kernel, a spike remains regardless of which CPU load type is being tested (idle or intentional load), likely due to other hardware interrupts, which will still halt the CPU.